Atlas Rover — System Requirements Specification

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1. Purpose

This document defines the approved system-level requirements for the Atlas inspection rover, a battery-powered tracked vehicle for warehouse inspection.

Revision note: bus voltage, CAN rate, temperature range and enclosure rating updated after the preliminary design review.

2. Power

R-101: The main power bus shall operate at 12 V nominal (approved).

R-102: The drive system shall support a maximum continuous bus current of 15 A.

R-103: The battery pack shall provide at least 4 hours of continuous operation (assumption: derived from customer interview; no battery test data yet).

3. Communications

R-201: All internal controllers shall communicate over CAN bus at 1 Mbit/s (approved).

R-202: Telemetry shall be sent to the base station over Wi-Fi 5 GHz.

4. Environmental

R-301: The rover shall operate across an ambient temperature range of -10 °C to 50 °C (approved).

R-302: The electronics enclosure shall be rated IP65.

R-303: Maximum vehicle mass is 18 kg.

R-304: Maximum vehicle speed shall be 1.5 m/s (approved).